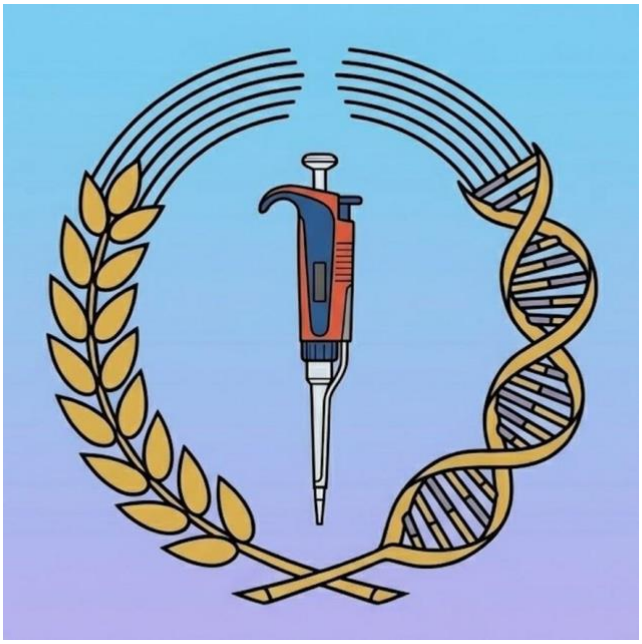
Cloner 64

User Handbook

A DNA Cloning Analysis App for macOS

Version 1.1 July 2026



Introduction

Cloner 64 is a DNA cloning analysis application for macOS. It replicates the look and ease of use of Serial Cloner, and runs natively on 64-bit Macs. Cloner 64 reads xdna and xprt files (Strider/Serial Cloner format) as well as FASTA, GenBank, APE, and SnapGene files.

If like me you depended on (and loved using) Christian Marcks's Strider and then Serial Cloner by Franck Perez, and were annoyed when Apple decided to no longer support 32-bit apps, then this app might be for you. There are other excellent DNA analysis apps out there for the Mac; however being a creature of habit, I prefer the Serial Cloner approach, and now that my old 32-bit Mac is falling to bits I decided to try and make a version of Serial Cloner that works on the new generation of Macs.

Cloner 64 is possibly not as elegantly crafted as Serial Cloner and does not cover all of the original features - for example there is no Golden Gate cloning, simply because it is not something that I ever did - too expensive! But there are a few things in this app that complement the original, such as predictive cloning analysis, shuttle vector route-finding, and context-aware methylation checking. The app was created using Anthropic's Claude and DeepSeek to assist with the coding and trouble-shooting.

This handbook covers the main features in Cloner 64 v1.1, to help you get started. Changes since v1.0 are listed under What's New in 1.1 below.

What's New in 1.1

Gel strength in the Virtual Cutter. An agarose slider from 0.5% to 2.0% repositions the bands to where they would run on a gel of that strength, and the co-migration warning now follows it, so you can find a gel that separates a troublesome pair before pouring one.

Residue readout on the hydropathy plot. Hovering the plot reports the residue number, its one-letter code and the hydropathy value at that point.

Type IIS enzymes are no longer offered in Predictive Cloning. BsaI, BsmBI, BbsI and SapI cut outside their recognition site, so their overhang depends on the target sequence and cannot be matched in advance. They remain available throughout the rest of the app.

Corrupt files are now reported on import. Characters in an XDNA or XPRT file that are not valid bases or amino acids used to be accepted silently; an alert now names them.

Corrections to reported positions and fragment sizes. Site positions in the Site Usage window and the Graphical Map are now numbered from 1, matching the Sequence Editor and the rest of the app — these previously read one lower. Digest predictions no longer show a spurious extra band when two enzymes cut at the same position on a circular molecule, which affected isoschizomer pairs such as MspI with HpaII.

Getting Started

Opening Cloner 64 for the First Time

Cloner 64 is not signed with an Apple Developer certificate, so macOS blocks it the first time you try to open it. This is a one-time step and the app works normally afterwards.

On macOS 15 (Sequoia) and macOS 26 (Tahoe) the app may appear to start — an icon shows in the Dock — but no window ever opens and no warning is displayed. That is macOS silently blocking it, not a fault in the app.

First try the simple route: move Cloner 64 to your Applications folder, right-click it and choose Open, then click Open again if a warning appears. On macOS 14 and earlier that is all that is needed.

If the app still will not open, Apple removed that shortcut in macOS 15 and one Terminal command is required. Open Terminal (Applications → Utilities → Terminal) and type the following, including the space at the end, without pressing Return: xattr -dr com.apple.quarantine

Then drag Cloner 64 from Finder into the Terminal window, which fills in the location for you, and press Return. If nothing is printed, it worked — open the app normally. Should you see "Operation not permitted", switch on Terminal under System Settings → Privacy & Security → App Management and repeat.

macOS tags every downloaded file with a quarantine marker. That command removes the marker from Cloner 64 only; it changes no security settings and affects no other application.

Opening Files

Cloner 64 supports the following file formats:

Format / Extensions	Notes
XDNA (.xdna)	Serial Cloner DNA format (read/write)
XPRT (.xprt)	Serial Cloner protein format (read/write)
FASTA (.fasta, .fa, .fna)	Plain sequence (read/write)
GenBank (.gb, .gbk)	Annotated sequence (read/write)
APE (.ape)	A Plasmid Editor format (read)
SnapGene (.dna)	SnapGene binary format (read)

To open a file, double click on it, or use File › Open or drag files onto the app icon in the Dock. Recently opened files appear under File › Open Recent. Ape and SnapGene files are detected automatically by their binary signature.

Creating a New Sequence

Choose File › New › DNA Sequence to create an empty sequence, or File › New › Protein Sequence for a protein. You can also paste a sequence from the clipboard using Edit › Paste into a new sequence window.

Saving and Exporting

Save your work with File › Save. Use File › Save As to save a copy with a new name. Export menus let you save DNA sequences as FASTA, GenBank, or XDNA, and protein sequences as FASTA or XPRT.

The Sequence Editor

The Sequence Editor is your main workspace. It displays the DNA sequence with line numbers, complementary strand for DNA, and colour-coded features.

Locked and Unlocked State

Sequences opened from disk are locked by default to prevent accidental edits. A “Locked” checkbox in the editor toolbar shows the current state. Uncheck it to enable editing.

Editing

Click anywhere in the sequence to position the cursor, then type to insert bases. Select a region by clicking and dragging, or use Shift+Click to extend a selection. Use the main Edit and Tools menus for copy, cut, paste, translation, and other options.

Features

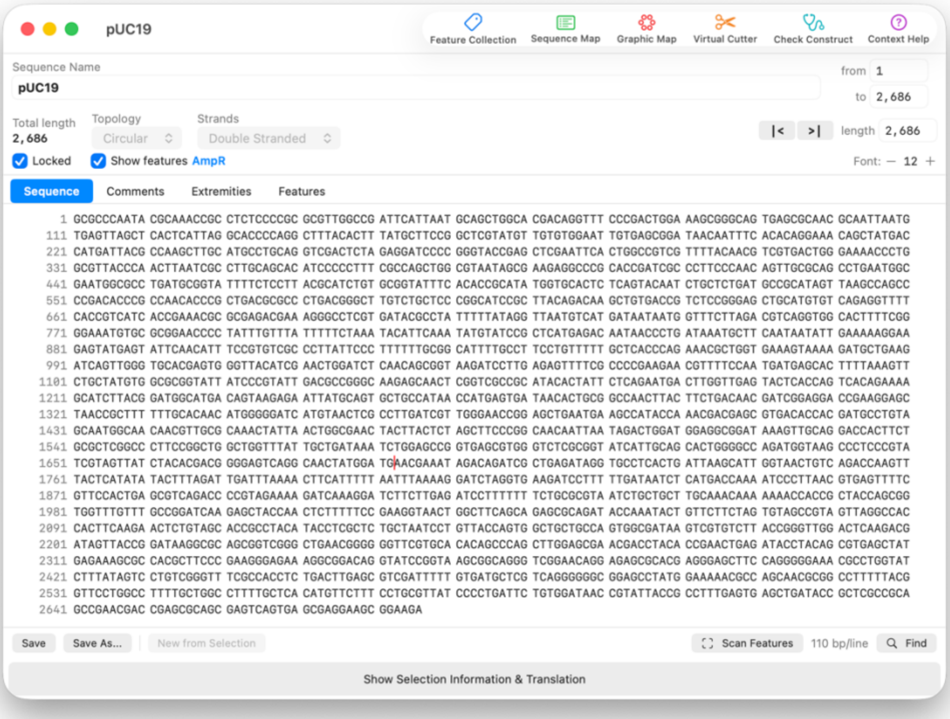
Annotated features (promoters, genes, origins, etc.) appear as coloured bars overlaid on the sequence. These are embedded in the .xdna sequence file or may be added from the Feature Collection by a search. Each feature has a name, type, location, and reading direction. From the Features tab you can edit individual features. Features that are added to the file are preserved upon saving the file.

Translation

Select a region of the sequence and choose Tools › Translate Selection to translate it into a protein sequence. The translated protein opens in its own window. Stop codons are shown as asterisks (*). Or click on “show selection information and translation”

Find Drawer

The Find drawer lets you search for specific sequences, restriction enzyme sites, and open reading frames (ORFs) within your sequence.



Sequence Editor View

Search for sequences, restriction sites, and ORFs opens on the right.

Translations and general information open underneath the sequence

Graphical Map

The Graphical Map displays your sequence as a circular or linear map, depending on sequence topology.

Restriction Enzyme Sites

Use the Sites toolbar button (Scissors) to choose which restriction sites to display:

- Unique Sites - enzymes that cut exactly once (cream-coloured)
- Double Sites - enzymes that cut exactly twice (brown-coloured)
- Blunt Sites - blunt-end cutters (teal-coloured)
- Particular Sites - choose specific enzymes from a list (blue)

Display Options

The Display button (Eye icon) lets you toggle features and ORFs, and selectively show or hide individual features and ORFs. The Split icon will give a view of the sequence below the map.

Methylation Sensitivity

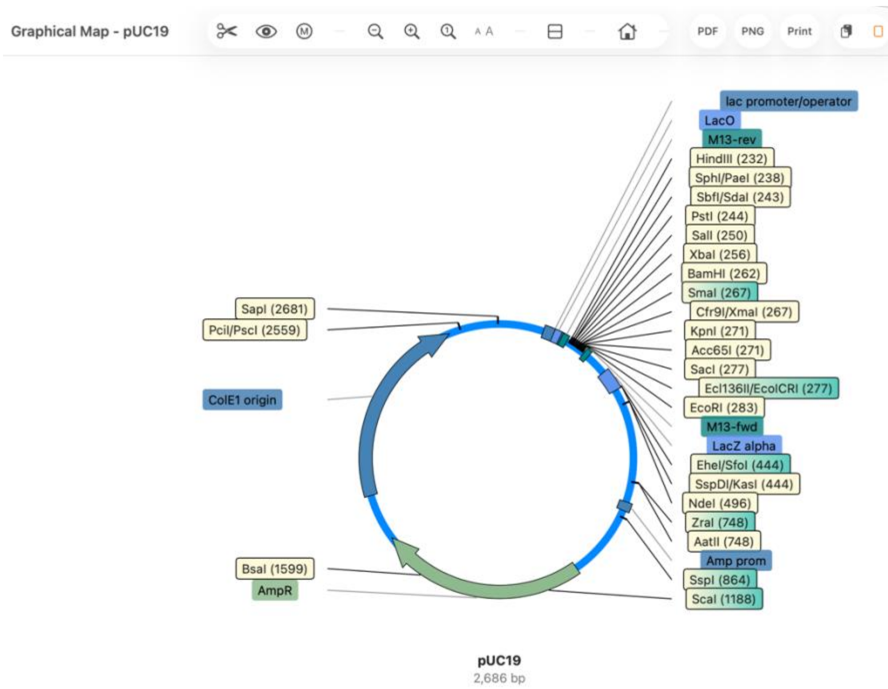
The methylation button (M icon) controls Dam, Dcm, and CpG methylation sensitivity. The app checks the sequence context at each cut site, so methylation warnings are only shown where the relevant methylation motif is present. Affected sites are shown in red on the map.

Zoom and Font Size

Use the magnifying glass buttons to zoom in and out. The text size buttons (A smaller / A larger) adjust label font size. Press the 1x button to reset zoom to 100%.

Export and Print

Export your map as PDF or PNG using the toolbar buttons. Use Print to send the map to a printer.



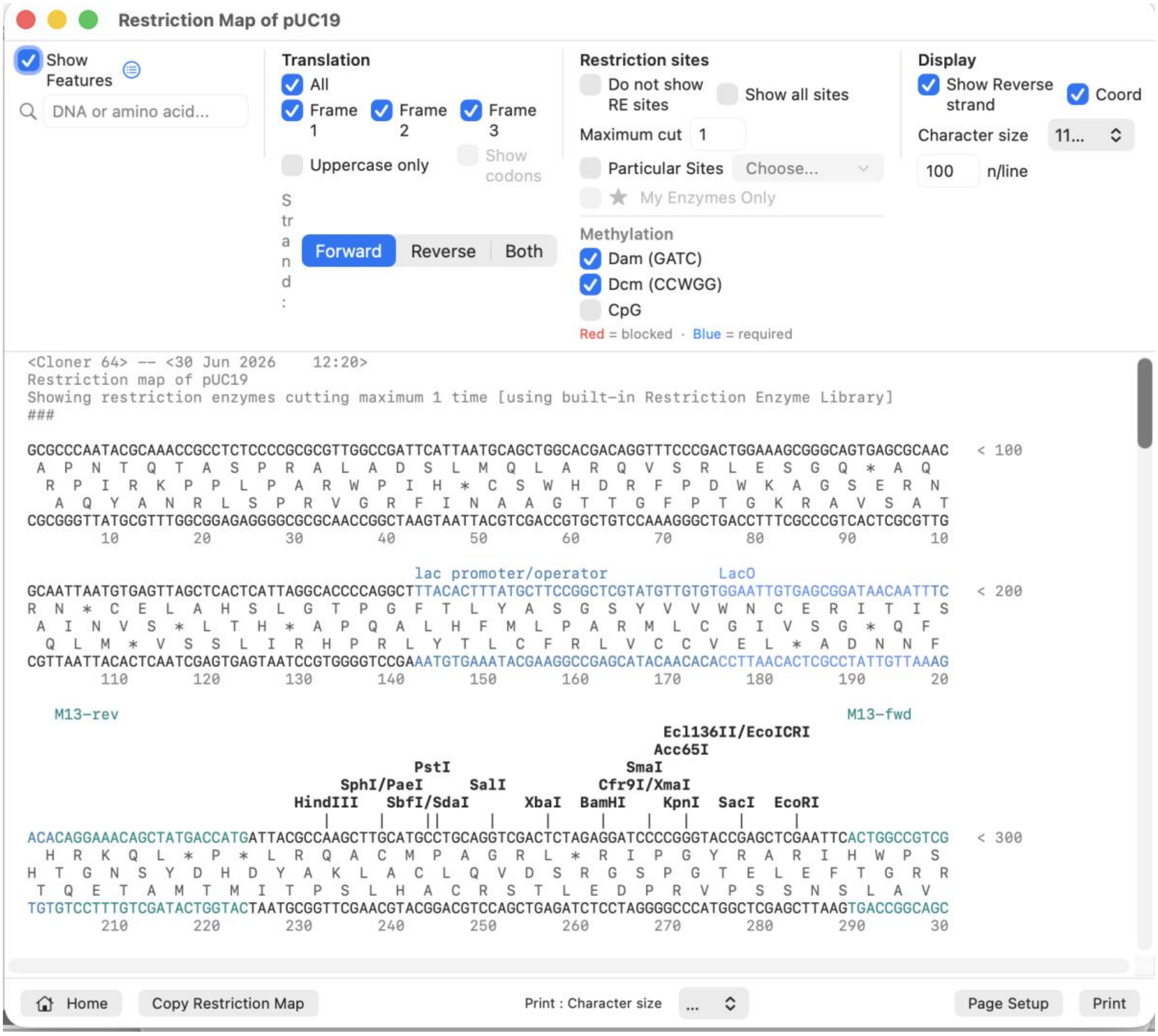
Graphic Map

Clicking on features allows you to select them.

Clicking on two restriction sites allows you to select the fragment(s)

Sequence Map

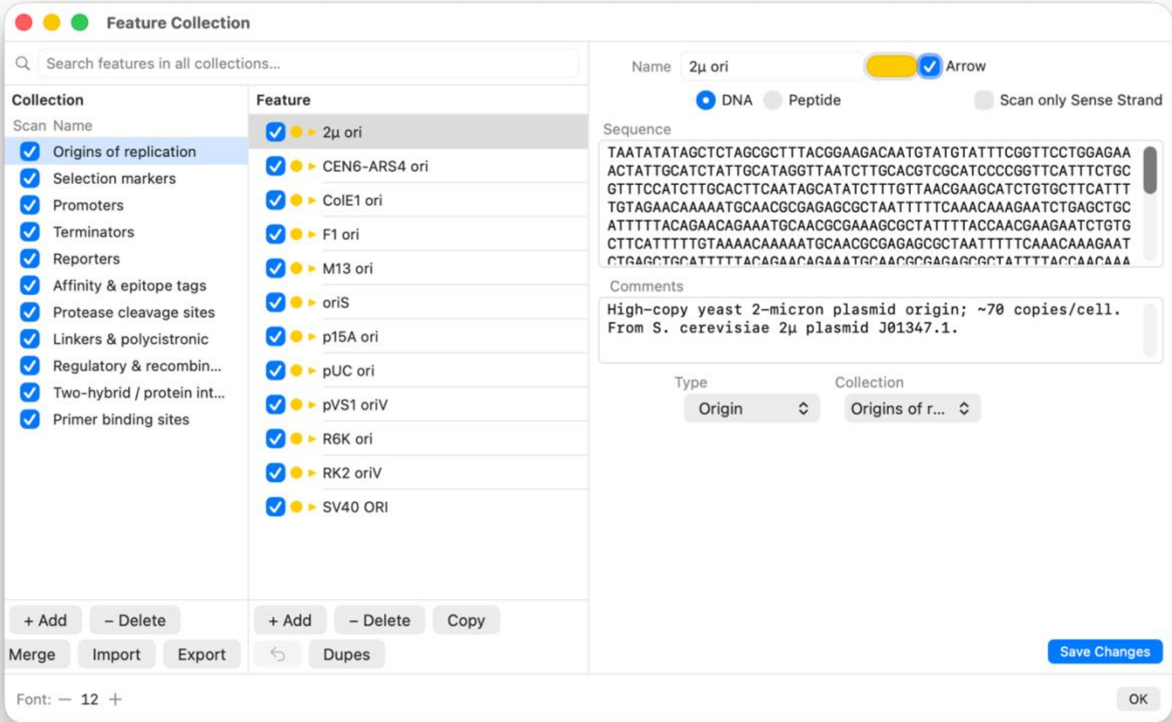
The Sequence Map (accessible from the Map icon in the Sequence Editor) displays a text-based restriction map of your sequence. Enzyme cut sites are marked directly on the sequence text, with optional translation frames for both forward and reverse reading frames shown above and below the DNA sequence. Feature annotations are displayed in context. You can configure the line width, and export the map via copy, or print.



Feature Collection

The Feature Collection (Tools › Feature Collection, or from the Label icon in the Sequence Editor) lets you view, edit, add, and remove annotated features (DNA or protein) which can be noted on your sequence by scanning. You can reorder features by dragging, and the collection can be screened for duplicates.

Use Tools › Scan Sequence for Features, or the Scan Features button below the sequence to automatically scan your sequence against the built-in Feature Library. The library contains common molecular biology elements organised into eleven collections — including origins of replication, selection markers, promoters, terminators, reporters, epitope tags, protease cleavage sites, and recombination sites. You can also create, import, and export your own feature collections.



Tools

Reverse Complement

Tools › Reverse Complement reverses the entire sequence and takes its complement (A↔T, G↔C). This is useful when you need to work with the opposite strand.

RNA/DNA Conversion

Convert between DNA and RNA using Tools › Convert to RNA (T→U) and Convert to DNA (U→T).

Site Usage

Tools › Site Usage opens a table showing all restriction enzyme recognition sites found in your sequence, including cut positions, fragment sizes, and whether each site is affected by methylation. You can filter by unique cutters, blunt cutters or absent cutters as well as restricting the list to your own enzyme stock.

Restriction Enzyme List

Tools › Restriction Enzyme List opens a searchable reference of all restriction enzymes in the database, showing recognition sequences, cut positions, overhang types, and methylation sensitivity. Isoschizomers with same recognition site and cut position are consolidated into combined entries (e.g. AflII/BspTI). You can add, edit, or remove enzymes.

My Enzymes

Within the Restriction Enzyme List you can earmark enzymes that you actually have in stock. Click the star (★) beside any enzyme to add it to your “My Enzymes” collection. Use the My Enzymes toggle in the toolbar to filter the list to only your starred enzymes. Your selection is saved between sessions. The My Enzymes filter carries through to other parts of the app: for example Check Construct can confine its digest suggestions to your starred enzymes only, saving you from being offered enzymes you don’t have.

Compatible Cohesive Ends

Tools › Compatible Cohesive Ends shows which enzyme pairs produce compatible sticky ends (e.g. BamHI and BglII), which is essential for planning cloning strategies. The table is computed dynamically from the current enzyme database, and can be searched by enzyme name or overhang site.

Cloning Vector Library

Tools › Cloning Vector Library opens the Shuttle Vector Library, which contains a metadata database (not the full sequence) of common vectors with their MCS enzyme information, selection markers, sizes, and categories. You can add your own vectors and import/export entries. Each vector in the library can be earmarked as one of yours using the star (★) button. Use the My Vectors toggle in the toolbar to filter the list to only your starred vectors. Your selection is saved between sessions and is used by the Predictive Cloning shuttle route search (see below).

Reference Tables

Quick reference tables for the standard genetic code and IUPAC nucleotide ambiguity codes are available under Tools › Genetic Code and Tools › IUPAC Nucleotide Codes.

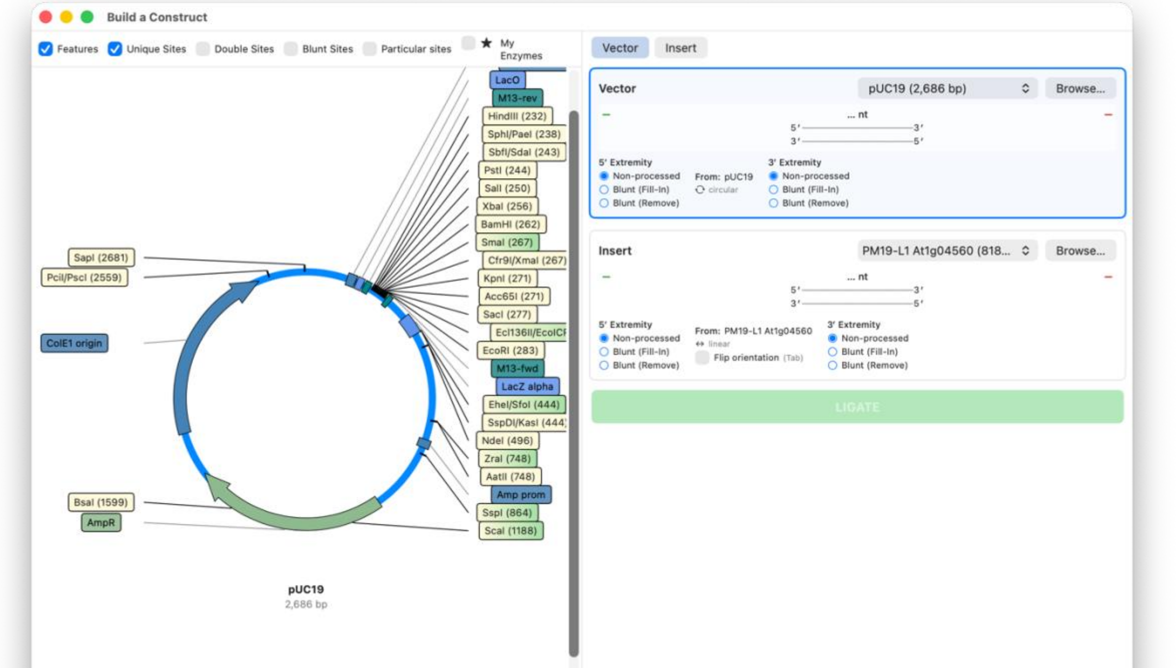
Functions

Build a Construct

Function › Build a Construct opens the Construct Builder for in silico ligation. Select a vector and insert, choose restriction enzyme cut sites on the graphical maps, and Cloner 64 will simulate the ligation, checking sticky-end compatibility and fragment orientation. The resulting construct can be saved as a new sequence.

Key features of the Construct Builder:

- Vector and Insert panels showing maps and details of 3’ and 5’ ends
- Sticky-end compatibility matching across enzymes
- Backbone and insert fragment choice using enzyme cut positions
- Overhang display allows end processing options (fill/trim)
- Handles both circular and linear sequences



Build a construct. Select restriction sites in the vector and in the insert, and press ligate. You can save the new construct or carry out a fresh ligation with different restriction sites.

Verify Construct

The Verify Construct function is made available by the Predictive Cloning and Construct Builder tools after a construct has been assembled. It provides a verification digest report for that specific construct, taking into account the insert position and length. Two modes are available:

Insert orientation must be determined - tick this when the ligation was non-directional and you need a digest that distinguishes forward from reverse insert. The tool looks for enzymes that cut the insert asymmetrically so the two orientations give different band patterns.

Orientation fixed - untick this when the cloning strategy already forces the orientation (for example, a directional two-enzyme ligation). The tool instead looks for flanking diagnostic digests that confirm insert presence.

Check Construct

Function › Check Construct or the Stethoscope icon in the Sequence Editor window opens the diagnostic digest advisor. Given a construct (or any sequence), it suggests restriction digest strategies to confirm that your plasmid is what you think it is or that your cloning worked. There are four modes:

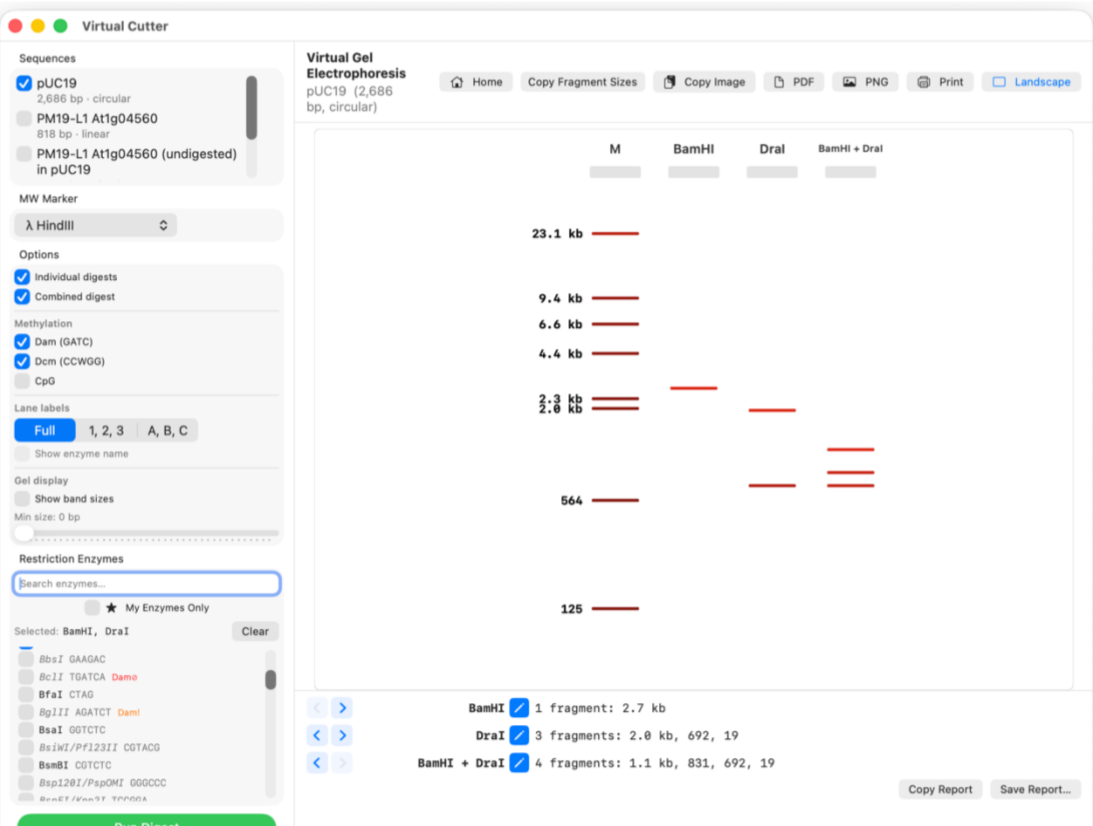
- Fingerprint mode - finds enzymes that give a distinctive gel pattern for the whole construct
- Presence mode - finds enzymes confirming that a feature is present
- Orientation mode - finds enzymes that can distinguish forward from reverse insert orientation
- Comparison mode to distinguish recombinants from parent vector

Virtual Cutter

Function › Virtual Cutter, or Scissors icon from the Sequence Editor, performs a virtual restriction digest. Choose one or more enzymes and the app simulates the digest, showing fragment sizes and a gel electrophoresis visualisation. Results can be exported as PDF, PNG, or printed.

Gel strength › The gel strength slider sets the agarose concentration, from 0.5% to 2.0%, and moves the bands to where they would run on a gel of that strength. Each concentration separates a particular size range well: lower percentages spread out large fragments, higher percentages spread out small ones. Fragments above the range bunch up near the well, and those below it run down near the dye front, so the same digest can look quite different on two gels. The caption under the slider shows the range the current setting resolves.

The co-migration warning on each lane follows the slider. Two fragments that merge into a single band on a 2% gel may separate cleanly on a 0.7% one, and if a different strength would resolve more bands a note appears suggesting it. This is a model based on standard agarose tables rather than a simulation of particular equipment: it is reliable about which bands separate from which, while exact spacing will depend on your buffer, voltage and run time.



Predictive Cloning

Function › Predictive Cloning opens the automated cloning strategy analyser. Select a vector and insert sequence, and the tool screens all enzyme combinations to find workable cloning strategies. Each strategy is scored on multiple criteria:

- Directionality - does the insert go in the correct orientation?
- Internal cuts - does the enzyme cut within the insert or vector backbone?
- Reading frame preservation - for fusion protein constructs, with per-junction frame offset control
- Methylation sensitivity   are any sites blocked by Dam, Dcm, or CpG methylation?

The tool supports partial digest strategies and compatible-end cross-enzyme cloning (e.g. using BamHI on the vector and BglII on the insert). You can define MCS regions with coordinate-based protected sequence boundaries, including origin-wrapping support for circular plasmids.

For each strategy, the results panel shows fragment sizes for ligation, and a per-strategy protocol export button. If a PCR route is chosen, the Design Primers button transfers the selected strategy to the Primer Design tool. The Build Construct button generates the predicted product as a new sequence with remapped features.

Type IIS enzymes › Enzymes that cut outside their recognition site — BsaI, BsmBI, BbsI and SapI — are not offered here. Every strategy this tool builds depends on knowing an enzyme’s overhang in advance so it can judge whether two cut ends will join, and a Type IIS overhang is whatever sequence happens to lie at the cut, a property of the target rather than of the enzyme. Golden Gate assembly, where that is the whole point, would need a different approach and is not currently supported. These enzymes remain available everywhere else in the app, including the Virtual Cutter and the site maps.

Insert Direction Preference

Use the Insert Direction control to specify whether the insert should go in the forward orientation, reverse orientation, or either. This narrows the results when orientation matters.

Fusion Cloning and Frame Alignment

When cloning a coding sequence as a fusion, enable Fusion Mode and select tag from the vector and ORF from the insert.

Multi-Source Scan

The Multi-Source Scan tab lets you search cloning strategies for an insert across all currently open sequences simultaneously, rather than against a single sequence. Useful when you have several candidate insert constructs for the same gene and want to find an easy strategy.

Shuttle Vector Routes

If you cannot easily find a strategy by using your chosen vector, the Shuttle Routes tab searches for PCR-free multi-step routes via intermediate vectors. The pathfinder finds a chain of compatible vectors to move the insert through step by step. As noted before, the Shuttle Vector Library, contains a metadata database (not the full sequence) of common vectors with their MCS enzyme information, selection markers, sizes, and categories. Thus once a strategy has been evaluated, this should be tested out using the full shuttle vector sequence.

Design PCR Primers

Function › Design PCR Primers opens the primer design tool. Select a template sequence, define your target region (or pick a feature/ORF), and the tool suggests forward and reverse primers with calculated melting temperatures and GC content. Primers can include 5′ tails for adding restriction sites. The visual template map has draggable handles for adjusting the target region. Designed primers can be saved and imported with tail/annealing annotations preserved. Primer-dimer screening is built in. Circular templates are supported.

Primer Stock

If you have a folder of existing primer files (.xdna format), you can nominate it as your Primer Stock folder in the Primer Design tool. The app will automatically screen your stock primers against the current template and, where a match is found, use the stock primer as a weighted candidate in the design results. This saves re-ordering primers you already have.

Run a PCR

Function › Run a PCR simulates a PCR reaction. Choose a template, provide forward and reverse primers, select a polymerase (Taq or Pfu), and the tool predicts the amplified product including any 5′ tails and Taq A-overhangs. The product can be saved as a new sequence.

Sequence Alignment

Function › Align Two DNA Sequences and Align Two Protein Sequences perform pairwise alignment of any two open sequences. DNA alignment uses word-based seeding with banded Needleman-Wunsch; protein alignment uses BLOSUM62 scoring with Smith-Waterman. Both show colour-coded matches, mismatches, and gaps with identity/similarity scores.

NCBI BLAST Search

Function › NCBI BLAST Search DNA and BLAST Search Protein open the NCBI BLAST web interface pre-loaded with your current sequence for online homology searching.

Hydropathy Plot

Function › Hydropathy Plot generates a Kyte-Doolittle hydropathy plot for the current protein sequence, useful for predicting transmembrane regions. The transmembrane threshold (1.6) is shown as a reference line. The plot is zoomable.

Moving the pointer across the plot shows a crosshair with the residue number, its one-letter code and the smoothed hydropathy value at that point. Note that the curve begins and ends short of the sequence itself: each point is the average of a window of residues centred on it, so with the default window of 9 the plot runs from residue 5 to the fifth residue from the end. Widening the window increases that margin, and the plot is directly comparable with ProtScale or EMBOSS output for the same protein and window.

Protein Sequences

Cloner 64 supports protein sequences with a dedicated viewer. Protein windows show the amino acid sequence with Clustal-style colour-coded residues and a legend. A properties panel shows molecular weight, isoelectric point (pI), and extinction coefficient. Proteins can be opened from XPRT files or created by translating a DNA selection.

Keyboard Shortcuts

Shortcut	Action	Shortcut	Action
Cmd+N	New DNA Sequence	Cmd+Shift+N	New Protein Sequence
Cmd+O	Open File	Cmd+S	Save
Cmd+Shift+S	Save As	Cmd+Z	Undo
Cmd+Shift+Z	Redo	Cmd+X/C/V	Cut / Copy / Paste
Cmd+Shift+V	Paste as New Sequence	Cmd+A	Select All
Cmd+U	Make Uppercase	Cmd+Shift+U	Make Lowercase
Cmd+T	Translate Selection	Cmd+L	Feature Collection
Cmd+B	Scan for Features	Cmd+Shift+K	Build a Construct
Cmd+Shift+D	Virtual Cutter	Cmd+Shift+R	Run a PCR
Cmd+Shift+A	Align Two DNA Sequences	Cmd+Shift+E	Export as FASTA
Cmd+P	Print	Cmd+Shift+P	Page Setup
Cmd+Shift+?	Show Help menu	Escape	Exit Full Screen

Tips and Tricks

- Use the Find drawer to locate restriction sites, ORFs, and specific sequence motifs.
- Double-click a feature on the graphical map to select it and copy its sequence to the clipboard
- When designing cloning strategies, check Tools › Compatible Cohesive Ends to find enzyme pairs that produce compatible sticky ends.
- Use Predictive Cloning to automatically screen all possible enzyme combinations for a vector/insert pair - it saves a lot of manual checking.
- The Primer Design tool can target a specific feature or ORF - use the target picker dropdown.
- PCR products from the PCR Simulation can be used directly as inserts in the Construct Builder.
- Methylation sensitivity is context-aware; the app checks the actual sequence context at each cut site rather than just flagging all sites for a sensitive enzyme.
- Star (★) your commonly used enzymes in the Restriction Enzyme List to build a My Enzymes set. Use the My Enzymes filter in Check Construct to get suggestions restricted to your own freezer stock.
- Star (★) the vectors you have in your lab in the Cloning Vector Library. Enable My Vectors Only in the Shuttle Routes tab to restrict the pathfinder to routes you can actually carry out.
- If you have a folder of existing primer files, nominate it as your Primer Stock in Primer Design. The app will automatically find and prefer matching primers you already have.
- The Shuttle Vector Library pathfinder can identify multi-step PCR-free routes between vectors that share compatible MCS sites.
- Context Help is available throughout the app. Turn on Context help from the main Help menu, then hover over any button, toggle, picker, or field for an explanation of what it does and when to use it.

Cloner 64 -Built for molecular biologists who miss Serial Cloner on modern Macs.